

REQUIREMENTS TABLE

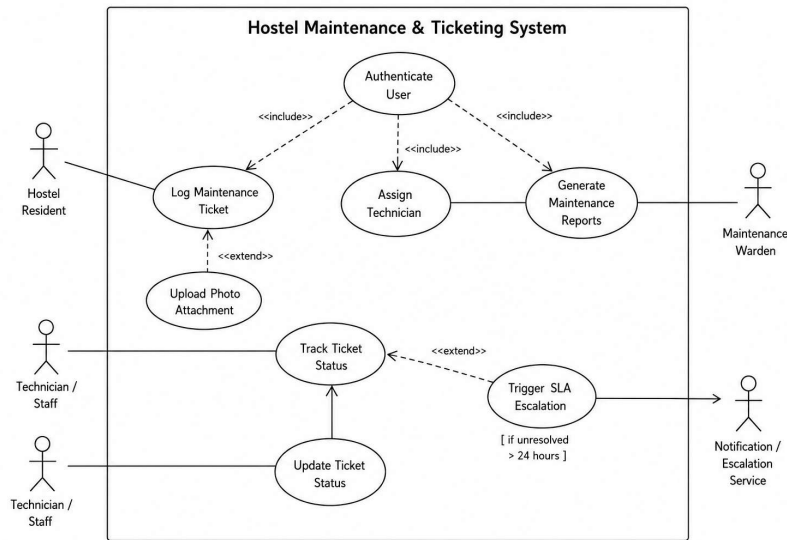
Problem Statement: Hostel Maintenance & Issue Ticketing System.

hostel administration requires a geo-tagged ticketing pipeline to report plumbing, electrical and carpentry issues, dispatch maintenance staff, track SLA timelines, and record resident closure sign-off.

Target stakeholders / Actor: Hostel Resident, Maintenance Warden.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The sys shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized (Plumbing, Electrical, Carpentry) issue severity.	High	Pass: Ticket is created with a unique ID and sent to warden. Fail: Form is submitted without room number or category.	Helps categorize problems and know the exact room location quickly.
FR-002	Functional	The sys shall authenticate users via institution credentials and restrict dashboard access based on roles (Resident, Warden, Technician).	High	Pass: User logs in and sees only their assigned dashboard. Fail: Student gets access to warden controls or wrong login works.	Protects student privacy and prevents unauthorized changes.
FR-003	Functional	The system updates and tracks ticket status (Open, Pending, In-Progress, Resolved, Closed) with timestamps.	High	Pass: Status changes correctly and records time of update. Fail: Status changes without saving time or skips required steps.	Helps students and wardens track repair progress in real time.
FR-004	Functional	The system shall send instant push and email notifications to residents and staff upon status transitions, technician	Medium	Pass: Alert is received within a few seconds of update.	Keeps everyone updated without needing to check the portal repeatedly.

		assignment, or schedule updates.		Fail: Ticket status updates but no alert is sent.	
FR-005	Functional	The system shall generate weekly/monthly reports for the Warden, detailing metrics such as recurring issues by room/block, average resolution time, and SLA compliance rates.	Medium	Pass: Warden can download summary report as PDF/Excel. Fail: Report does not load or shows wrong calculated data.	Helps the hostel team find recurring problems and track worker performance.
NFR-001	Non-Functional (Performance & Security)	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.	High	Pass: Auto-alert goes to chief warden if no action after 24 hours. Fail: 24 hours pass and no alert is triggered	Ensures emergency problems are not forgotten or delayed.
NFR-002	Non-Functional (Availability & Reliability)	The system shall maintain an uptime of at least 99.5% during operational hours (24/7), excluding scheduled maintenance windows.	High	Pass: App runs smoothly with almost zero unplanned downtime. Fail: Portal crashes or is unavailable during regular hours.	Students need to report emergency hostel issues at any time of day or night.



Use-Case Flow Specification

Use Case: Log Maintenance Ticket

- **Use Case ID:** UC-01
- **Primary Actor:** Hostel Resident
- **Preconditions:** The resident is registered in the hostel database and has valid college login credentials.
- **Postconditions:** A new maintenance ticket is logged with a unique Ticket ID, marked as Open, and appears on the Warden's dashboard.

Main Success Scenario:

1. Resident opens the portal and selects **Create New Ticket**.
2. System includes Login and validates the resident's session.
3. System displays the ticket submission form.
4. Resident selects the room number, block, and issue category (Plumbing, Electrical, Carpentry).
5. Resident enters a brief description of the issue.
6. Resident optionally extends the flow to attach a photo of the damaged item.
7. Resident clicks **Submit**.
8. System validates required fields, creates a unique Ticket ID, timestamps the entry, and sets status to Open.
9. System shows a confirmation message to the resident with the Ticket ID.

Alternate Flow (Missing Required Information):

- **4a.** Resident leaves the room number or category field blank.

- **4b.** Resident clicks **Submit**.
- **4c.** System detects missing mandatory fields, blocks ticket creation, and displays an error message: *"Room number and category are mandatory."*
- **4d.** Resident fills the missing fields and resubmits (returns to Step 7).